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Pushing Boundaries with Zero Flaring Solutions Utilizing the Multiphase Pump to Unlock Constrained Production in a High H₂S Offshore Field

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Abstract

We often think of flow assurance as subsurface or downhole-related issues. Still, in many cases, the main challenge is the topside: the back pressure of the production line or sea line imposed by the surface facility, or the design operating pressure that highly limits what the well can deliver. To work around this, ADNOC Offshore, in collaboration with its business partner, trialed a Multiphase Pump (MPP) to reduce the wellhead pressure and stabilize the flow below the existing back pressure. Several challenges were addressed during the planning phase, including the selection of pilot wells, pump specifications, job modeling, surface footprint, and integration with the existing facility. Finally, over several days, the well was tested at different flow rates, with WHPs as low as 269 psig, while maintaining full online operation against the 675.9 psi production line pressure. The production increase was immediate and sustained: oil rates went from 1,477 to over 2,300 BOPD, and gas rates more than doubled. Managed to achieve the highest production figures from existing inactive wells without subsurface interventions while increasing production to meet the field target, while maintaining zero flaring during the entire operation. This success wasn't just about flow rates. It opened the door to bringing marginal and low-pressure wells online through existing facilities—wells that would otherwise likely have been written off. With minimal modification to infrastructure, the MPP permitted to break through a surface bottleneck that's all too common in mature offshore assets. In high H₂S offshore environments, particularly for marginal or constrained wells, managing the issue is exceptionally challenging due to the limited availability of injection lines. Traditional strategies often require flaring during cleanup or stimulation, which contradicts sustainability goals and regulatory standards. The future of the energy transition lies in innovation and decarbonization. This paper addresses this issue by presenting a zero-flaring approach using Multiphase Pumping technology, which holds promise for extending well life and reducing environmental impact.

Introduction

Hydrocarbon production and processing, when conducted responsibly, are a cornerstone of the energy transition, ensuring the global demand for reliable and safe energy.¹ ADNOC has declared its strategy to increase crude oil production capacity to 5 million barrels per day by 2027, achieve UAE gas self-sufficiency, and produce 1 billion cubic feet per day of unconventional gas², while reducing GHG emissions intensity by 25% (scope 1 and scope 2) by 2030 and targeting to achieve Net Zero by 2045, through multiple strategies, including achieving greater energy efficiency, zero routine flaring, and reductions in methane emissions³.

One of the adopted strategies to achieve these two seemingly conflicting objectives is described herein and involves extending the production life of already existing inactive wells. This approach hinders WHP production against the sealine and necessitates recurring restarts due to liquid loading, without resorting to flaring operations.

Flow assurance is the ability to produce hydrocarbon mixtures reliably, economically, and safely from the reservoir to the sales point over the life of a field, in any environment.⁴ Production bottlenecks can arise both downhole and at the surface. Downhole constraints—such as skin effects, frictional pressure losses, formation damage, and scale—reduce fluid inflow to the wellbore. At the surface, limitations occur within equipment and flowlines—such as wellhead chokes, separator capacity, gathering system pressure, and pipeline backpressure—that can restrict production rates despite adequate reservoir deliverability (Economides et al., 2013)⁵. In mature offshore assets, the latter is all too common. High line pressure may hinder the applicability of re-injection to existing production lines, either because of the inability to flow well at wellhead pressure above line pressure or due to pressure rating limitations of the surface equipment.⁶ Gas flaring, particularly in high H₂S offshore environments, presents not only operational and environmental challenges but also significant regulatory constraints. In Abu Dhabi, flaring is strictly governed by Law No. 8 of 1978 on the Conservation of Petroleum Resources. This law mandates that associated gas be utilized, reinjected, or otherwise managed in a manner that avoids atmospheric release unless explicit government authorization is obtained. Routine flaring—especially in the case of sour gas containing toxic H₂S—is further restricted under Law No. 24 of 1999 for Environmental Protection, due to the harmful impact on air quality and public health. These regulations are reinforced by ADNOC's corporate alignment with the "World Bank's Zero Routine Flaring by 2030 initiative" and its internal decarbonization targets. The successful deployment of the Multiphase Pump (MPP) in this trial—facilitating production without surface flaring—demonstrates both technical innovation and full compliance with local environmental laws and ADNOC's sustainability commitments, while avoiding the legal and reputational risks associated with sour gas flaring in sensitive offshore fields.

Previous studies have shown that reducing surface system backpressure—particularly at the wellhead—can significantly improve production from constrained wells. Modeling work by Economides et al. (2002) has demonstrated that enhanced interaction between the reservoir, well, and surface pump systems can mitigate flow restrictions caused by high surface backpressure, allowing for increased inflow without requiring subsurface intervention.⁷ In that, utilizing multiphase pumps is a viable solution to enable a flaring-free option, temporary production wellhead pressure reduction, with the result of production increase, to maintain a stable production from wells otherwise limited by the production line pressures⁸.

Field Background

All candidate wells are located within the same oil field operated by ADNOC Offshore. This field was discovered in the 1960s, with the final investment decision (FID) made in the early 2000s, and production started shortly thereafter. The wells draw from the Upper Jurassic section of the Arab Formation and are managed via artificial islands in shallow waters in the UAE.

The Arab Formation features a carbonate ramp system with subtidal to supratidal facies, exhibiting a regressive, shallowing-upward depositional cycle. It consists of shallow-water carbonates, including limestones, dolomitized limestones, and dolomites, with minor amounts of anhydrite. The formation exhibits heterogeneous micro- and macropore systems, with grain-supported carbonates displaying bimodal pore networks, and dolomite streaks often improving permeability. The fluid produced contains both oil and sour gas. Its H₂S content can correlate with anhydrite thickness and burial maturation, varying between 1% and 25%. The GOR of the wells in the field is expected to remain between 500 and 1000 scf/bbl.

The main objective of the scope was to assess the ability of a temporary well-test setup, with a Multiphase Pump (MPP) to reactivate an inactive well, and maintain and increase its production rate while pushing down the wellhead pressure and stabilize flow below the existing back pressure for a well in a mature field, with a high GVF and high H₂S content, for an extended period. The operator wanted to assess the maximum deliverability and the maximum safe drawdown for the well in question.

Application of MPP for Pressure Boosting and Production Sustainability

Multiphase Pumping is essentially a means of adding energy to the unprocessed effluent, which enables the liquid/gas mixture to be transported over long distances without the need for prior separation or flaring.⁹ Multiphase production schemes present significant advantages over single-phase ones, including (but not limited to) substantially reducing the flowing wellhead pressure by adding incremental head to the raw stream to overcome the backpressure from downstream pipelines and facilities, and improving the possibility of zero gas flaring.¹⁰ Multiphase boosting has also proven to be an effective tool for cost-effective enhanced oil and gas production.¹¹

Using an MPP to manage untreated crude offers significant technical and economic advantages over the conventional arrangement of a gas compressor paired with a high-pressure pump. This integrated design reduces footprint and weight, simplifies piping and controls, and minimizes operating expenditures. In addition, multiphase pumps are engineered to tolerate wide variations in gas-volume fraction and can handle transient flow regimes, including slug flow, without performance loss or shutdowns. By using a single driver to boost both phases simultaneously, they also achieve higher energy efficiency and operational reliability compared to the combined compressor-pump setup, which has more components, higher power consumption, and greater maintenance requirements.¹² This flexibility makes it a suitable solution for mobile and temporary setups, where temporary pressure boosting is required, during well testing or while a permanent solution is being designed or installed. In this study, the temporary surface well testing layout was integrated, enabling an assessment of well deliverability and informing future decisions by ADNOC on permanent installations, upgrades, or alterations.

The MPP was selected, and the system was configured to meet the specific requirements of this well. It was then integrated into the existing surface system.

Rather than a stand-alone asset, the MPP was a part of a Pressure boosting system that included a Multiphase Flow Meter (MPFM) (enabling continuous monitoring of flow rates, in-line condition, Gas Volume Fraction (GVF), and the tracking of water volume recirculation through the pump) and a Well Test Separator. The Well Test Separator was used to establish the natural flow baseline when the WHFP was above the pipeline injection pressure (when the boosting system was bypassed). It was then maintained as a recirculation vessel, allowing for the recirculation of produced liquids to facilitate GVF maintenance when required. [Figure 1](#) below shows a simplified process flow diagram (PFD) of the boosting station integrated into the existing Surface Well Testing setup that was present in the facility.

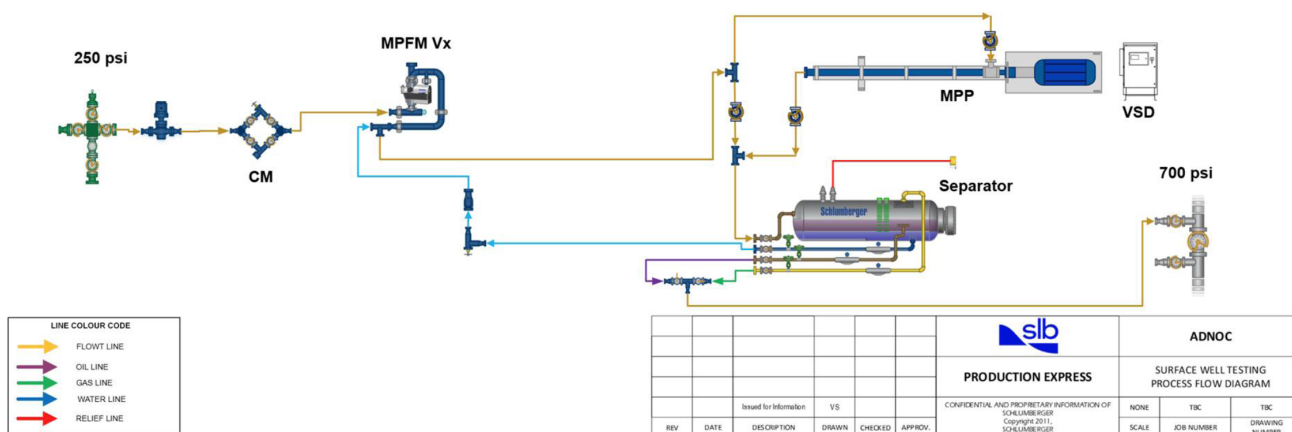


Figure 1—A simplified PFD of the proposed boosting setup used for the MPP trial in the selected well.

Figure 2 illustrates a general diagram of the pressure behavior expected across the MPP. Note that the anticipated pressure loss due to friction during transportation must be accounted for when defining the minimum pump discharge pressure. The multiple possible pressure variations (ΔP) throughout a process facility with a boosting station are represented in Figure 2 below.

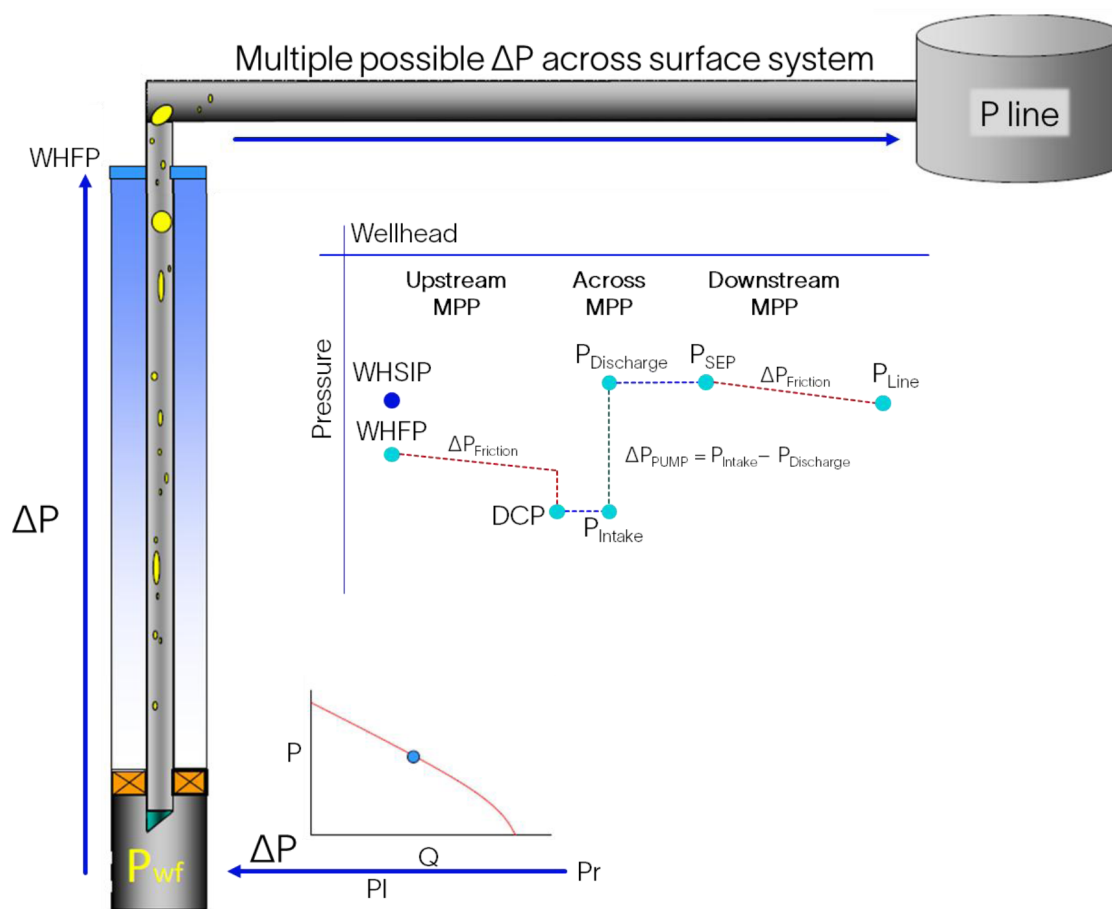


Figure 2—Expected pressure behavior of a surface production system including a separator and an MPP (Upstream SEP)

Modelling and Candidate Selection

The candidate selection is paramount for the success of any trials, as it needs to be ideal to showcase the solution to its full extent. For this trial, the well selection analyzed potential wells, reviewing well test data,

the latest available reservoir pressure, current production conditions, and reservoir fluid PVT data, using the most up-to-date data available for the wells.

To perform the design, a steady-state multiphase flow simulator was used to optimize production from the reservoir to the processing facility, encompassing both the wellbore and surface systems. It simulated pressure behavior throughout the pipeline and at all pressure and temperature transition points. A nodal analysis was performed for each required WHFP targeted, and the expected liquid flow rate was determined based on the known reservoir characteristics. The minimum discharge pressure for the pump was determined based on the reported injection point pressure. Considerations were made for any expected pressure losses due to friction between the MPP discharge and the injection point, based on the facility P&ID.

Once the expected production rates per WHP were determined, the results were used as entry values for sizing the MPP and estimating the volume to be handled by the pump, taking into account the required discharge pressure.

Table 1 below presents the contour conditions used for the MPP dimensioning, confirming that the boosting setup can safely and continuously manage the planned WHFP. The design boundary considered the minimum expected pressure boost between the MPP inlet and discharge, as well as the minimum volumes of liquid and gas to be handled, based on the predicted flow rates produced by the well.

Table 1—Simulation input for the volumes and pressures to be handled by MPP. Results from Steady State simulation, with a safety margin added. Gas rate calculated based on liquid rate simulation and fluid GOR.

Case	Target WHFP	MPP Pressure intake	Expected Liquid Rate	Expected Gas Rate	Minimum Boost pressure	MPP Pressure discharge
	[psig]	[psig]	[bbl/d]	[MMscf/d]	[psig]	[psig]
1	250	200	2850	1.85	550	750
2	350	300	2400	1.56	450	750
3	450	400	1900	1.23	350	750

MPP Characteristics

The selected MPP for this test was a horizontal multistage surface pump (HSP) due to its Field-proven REDA Multiphase Gas-Handling (MGH) - Helico-Axial Pump (HAP) technology. The decision was made based on the MPP's capacity to handle high GVF (up to 90% nominal) and its wide operational range. Table 2 below shows the operating specifications of the chosen MPP, while Figure 3 shows the pump picture (courtesy of SLB).

Table 2—Specification of the chosen MPP HPS unit selected for the operation.

Multiphase HPS Specification	
Volumetric rate actual, bpd [m ³ /d]	3,150 to 50,000 [500 to 8,000]
Liquid rate at 75% GVF, bpd [m ³ /d]	700 to 11,400 [110 to 1800]
GVF capacity	Up to 90% (for continuous operation)
Max. pump boost pressure, psi [bar]	900 [60]
Max. working pressure, psi [bar]	2,000 [135]
Unit integrity pressure, psi [bar]	3,000 [210]
Process fluid temperature, °F [°C]	60 to 175 [15 to 80]
Total suspended solids (TSS), ppm	<500
Ambient Temperature, °F [°C]	-40 to 104 [-40 to 40]

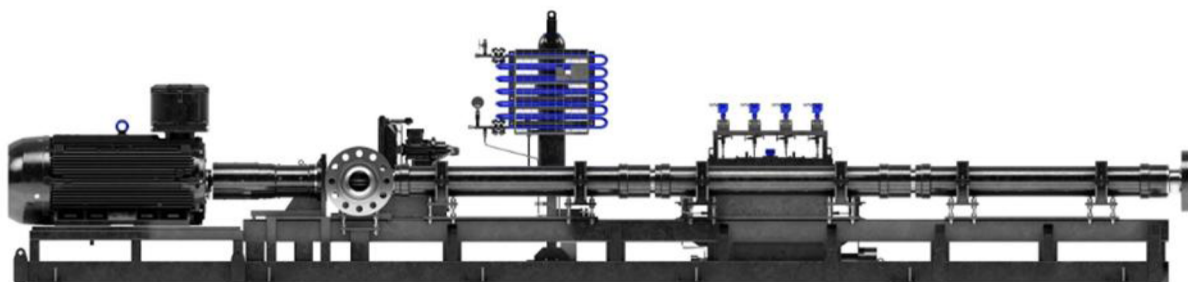


Figure 3—Multiphase pump model utilized in the described setup (courtesy of SLB)

This model was chosen based on its extensive field test track, highly flexible operating envelope, ease of installation, and practical modularity – all characteristics that make it an ideal temporary, mobile solution.

Multiphase Flowmeter (MPFM) Characteristics

Continuous evaluation in real time—including gas production rates, water cuts, and diluent rates—is critical to improving reliability and meeting production plans. Multiphase meters provide these real-time phase rates to underpin control decisions.¹³ To enable continuous control of the process parameters, the design incorporated an inline Multiphase flowmeter. The meter chosen allows for great flexibility in fluid parameters and is not affected by the well flow conditions, making it ideal for this study. The MPFM was sized based on the line conditions expected from the available information and the simulations performed, and it served as the primary metering device during the periods of boosted flow through the MPP. The working parameters of the MPFM selected for the study are presented in Table 3, indicating that it covered the operational envelope of the well and extended beyond.

Table 3—Specification of the chosen MPFM unit selected for the operation.

MPFM Specification	
Service	Sour (as per NACE MR0175 / ISO 15156)
Max. working pressure	5,000 psi [34.5 MPa]
Design temperature	–50 to 250 °F [–46 to 121 °C]
Electronics temperature	–40 to 185 °F [–40 to 85 °C]
Water/liquid ratio	0 to 100 %
Gas volume fraction	0 to 100 %
Liquid viscosity at line conditions	0.1 to 2,000 cP [0.0001 to 2 Pa·s]
Repeatability (total mass rate)	Better than 1 %
Max. liquid flow capacity	4,000 to 100,000 bbl/d (for the chosen size)
Max. gas flow at 10 MPa	11 to 250 MMscf/d (for the chosen size)
Repeatability (total mass rate at line conditions)	Better than 1 %
Resolution (total mass rate at line conditions)	Better than 0.1 %

Field Execution and Results

The scope of work included selecting the well based on available data before it became inactive. This was followed by designing the integration of the MPP, which involved setting up the existing surface equipment on the pad and connecting it with the current well test system. The next step was to size and choose the appropriate MPP model, as well as to design the expected surface setup to connect it with the existing well test facilities in the area. The program for the well involved restarting it with the MPP, allowing for a natural

flow at the lowest possible WHP to keep the separator pressure above the seal line. After establishing the natural flow rate behavior at around 700 psi WHP, the MPP was engaged to increase pressure from the intake line, connected to the well, to the discharge line, which runs through the injection line. The well's flow rate was measured using a Multiphase Flow Meter (MPFM). During the boosted periods (MP#2 to MP#4), the SEP was kept engaged as a workaround to recirculate liquids (water or condensate) as needed to keep the GVF of intake fluid within the MPP's specifications.

Period MP#1 aimed to establish a baseline (if possible) and determine the maximum natural production rate of the well, taking into account the restrictions imposed by the seal line backpressure. The subsequent periods were monitored to assess the increase in production. Additionally, reductions in WHP were planned to evaluate the well's maximum deliverability. Once the equipment was installed according to the P&ID and pressure and function tests were performed per the HAZOP, the well—initially shut in with a WHSIP of 930 psia—was opened in two phases: Phase #1 involved natural flowback via the separator, and Phase #2 involved boosted flowback via the MPP.

Table 4 below presents the actual sequence of WHP used to assess the MPP's capacity to achieve and sustain well production against the injection line. Figure 4 below provides an overview of the flow behavior, illustrating the observed pressure and flow rate during the mentioned periods.

Table 4—Planned sequence of flow and metering periods, considering the desired WHP and MPP discharge pressure requirement for reinjection.

Flow Period	WHP [psig]	Production Type	MPP Intake Pressure [psig]	MPP Required Discharge Pressure [psig]	Sealine Pressure [psig]
MP#1	700	Natural	—	—	656 psi
MP#2	450	Boosted by MPP	292	> 660 psi	
MP#3	350		288		
MP#4	250		179		

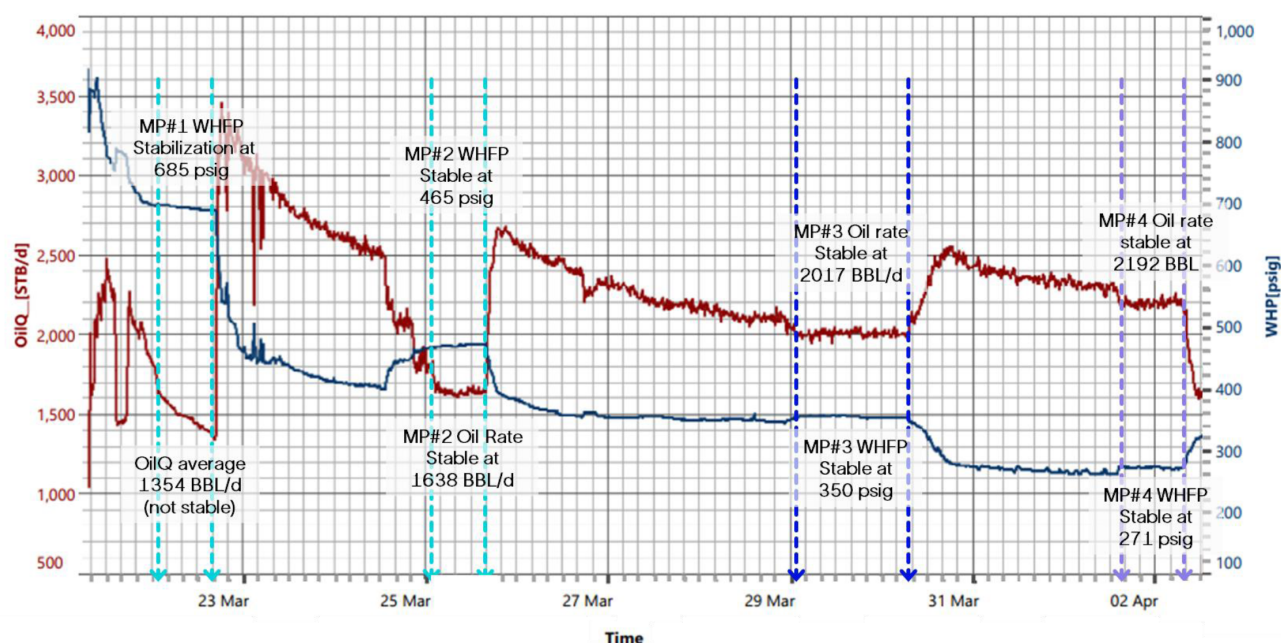


Figure 4—Representation of the results for WHP versus oil flow rate for each of the MPs described above. Note the difference in stability of the flow rates for each stable WHP between the periods with the boosting station and the period with natural flow.

Phase #1: Natural Flowback via Separator

MP#1 – Natural Flowback

The well was aligned through to the surface choke manifold and opened aligned through the MPFM and the SEP, bypassing the MPP system, to establish a natural flow back baseline. The production choke was gradually increased through choke size management to control the WHP and SEP pressures, ensuring they remained above the injection line pressure and guaranteeing zero flaring operations. The flow rate behaviour was monitored for a stable WHP well, which was kept in this condition for at least 3 days, with an expected WHFP of 650-700 psi. As observed in Figure 4 above, the pressure remained stable (likely due to the surface line backpressure), while the well flow rate never reached stability during the period. The baseline flow for this period was defined as the average flow rate for the last 8 hours of natural flow and was calculated to be 1462 barrels per day (BBL/d) of oil and 0.779 million cubic feet per day (MMscf/d) of gas.

Flow back data was continuously monitored and periodically assessed. Once the objective in this stage was achieved, the flow transitioned into Phase #2, but not before calibrating the MPFM parameters and collecting fluid references.

Phase #2: Boosted Flowback via MPP

MP#2 – Boosted flow WHP targeted at 450 psig. The fluid pathway was aligned to run through MPFM, MPP, SEP, and into the sea line in a closed loop. The choke size was adjusted until WHFP reached the target of 450 psig. After reaching the flow conditions set for MP#2, the well flow rate was monitored until it stabilized, then observed. The flow rates were measured and tracked throughout the entire period, and the well's deliverability at this pressure corresponded to the stable flow rate. All other process parameters, such as MPP intake and discharge, and injection line pressure, were also monitored and recorded. The results during the MP#2 period showed not only a decrease in WHFP with an increase in flow rate but also a stabilization of flow behavior for oil and gas, WHFP dropped from 685 psig to 465 psig, and oil production increased to 1671 barrels per day (BBL/d), while gas rates rose to 1 million cubic feet per day (MMscf/d).

MP#3 – Boosted flow WHP targeted at 350 psig. Once MP#2 was validated, the choke size and MPP frequency were adjusted to enter Flow Period MP#3, with a target WHFP of 350 psig. At this stage, note was made that the choke size was no longer leading the WHFP variation. The flow rates were then measured and monitored for the entire period, and the well deliverability at this pressure corresponded to the stable flow rate period. All other process parameters, such as MPP intake and discharge, and injection line pressure, were also monitored and recorded. Again, we observed that the WHFP was reduced to 350 psig, accompanied by a further increase in oil and gas flow rates, which stabilized at 2,114 barrels per day (BBL/d) and 1.269 million cubic feet per day (MMscf/d), respectively.

MP#4 – Boosted flow WHP targeted at 250 psig. By the end of Flow Period MP#3, the MPP settings were adjusted once more to achieve a WHFP of 250 psig, as required by the MP#4 plan. Once again, the choke size was no longer leading the WHP variation, and the MPP frequency had to be adjusted to achieve the target of 250 psi WHFP. We completed a WHFP of 270 psig, and the oil rate subsequently increased to 2,323 barrels per day (BBL/d) after stabilization, while the gas rate rose to 1.559 million cubic feet per day (MMscf/d). All other process parameters, such as MPP intake and discharge, and injection line pressure, were also monitored and recorded.

At the end of these periods, the WHFP was returned to the one in MP#1, and we observed that the well was unable to sustain natural flow. The boosting station was disconnected and removed from the island to be moved to the next test pad.

Throughout all the flow conditions enhanced by the MPP, the discharge pressure was maintained above 660 psig, which was required to overcome the injection line pressure. The minimum pump intake pressure

was 179 psig during MP#4, and this was the moment at which the maximum pressure boost was required from the MPP, which was 491 psig, to meet the requirement of surpassing the injection line pressure.

The volume of both oil and gas at line conditions was managed by the MPP as expected. Table 5 below summarizes the actual flow rates and pressures at the critical points of the surface production system, from the WHFP to the injection line.

Table 5—Table captions should begin with a short description of the table. Format tables using Microsoft Word table commands and structures. Do not create tables using spaces or tab characters. Large tables presenting rich data should be given as separate Excel or .csv files, not as part of the main text.

WHP [Psi]	Chock Size [1/64]	Pump Intake Pressure [Psi]	Liquid Rate [bbl/d]	Gas Rate [MMscfd]	GOR @STD [Scf/bbl]	Boost pressure [Psi]	Pump discharge pressure [Psi]
689	59	—	1462	0.779	527	—	—
450	40	292	1671	1.000	601	369	662
350	80-72	288	2114	1.269	601	374	663
250	128	179	2323	1.559	671	491	671

With the data from the three stable flow periods (MP#2, MP#3, MP#4), boosted by the MPP, and utilizing a steady-state multiphase flow simulator it was possible to create a model for this well's production profile and assess multiple operational conditions to support future decisions on the reintegration of this well into production. Figure 5 below shows the nodal analysis results for the model (which used an average PI between the three periods). This further supports the value added by the MPP boosting setup, which aligns with the expected behavior based on the WHFP required for the Pressure Boosting solution validation.

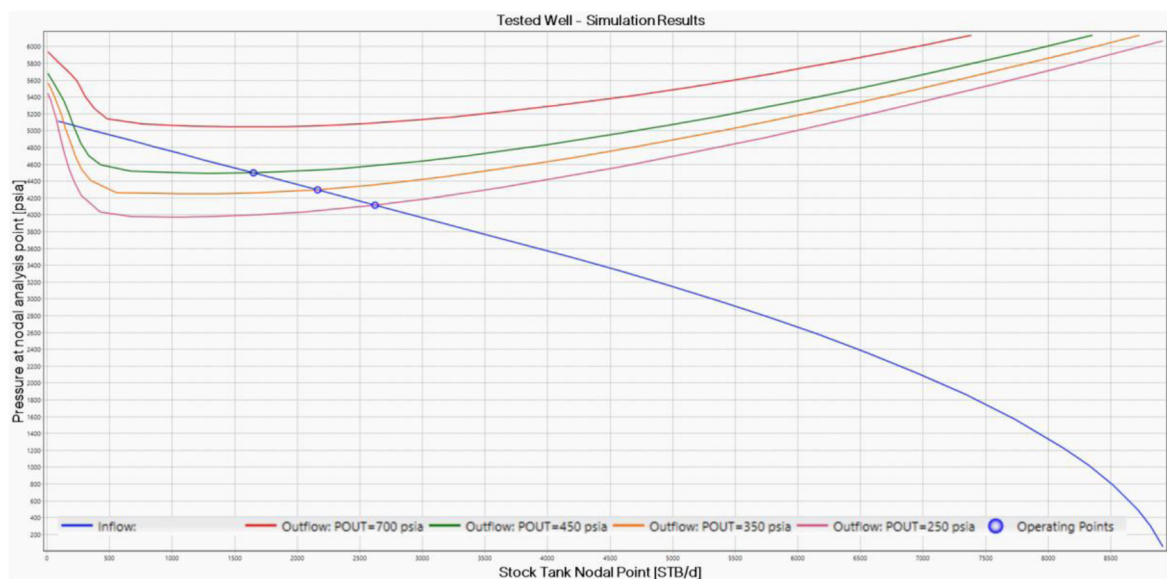


Figure 5—Model for the inflow and outflow of the tested well. Nodal analysis indicates that a wellhead pressure of 700 psia (WHP > production pipeline pressure) yields unconverging results, representing unstable flow conditions that are likely to lead to a stoppage of the well from flowing and a natural shut-in of the production with time.

During the total duration of the flow of the well, it produced 23,738.68 bbl of oil and 14.60 MMscf of gas. When considering the total volume produced from well opening to shut-in, the cumulative volumes were over 30,000 barrels of oil and 18.9 MMscf of gas. The successful incorporation of the boosting station enabled the entire test to be completed without gas flaring, which resulted in the emission of 1,000 tons of CO₂e (exclusively from the gas).

A projection of yearly production under WHFP of 250 psi corresponds to 847,895 barrels of liquid and 569.04 million Cubic Feet of gas per year. These values enable the operator to assess the viability and the next steps in reintegrating this well into production.

Conclusion

The trial has conclusively demonstrated the exceptional capabilities of this innovative technology, marking a significant milestone in our operations. Successfully connecting the boosting station reactivated the well and maintained stable production at multiple WHFPs, even when pressures were up to 450 psi below the injection line pressure—an impressive feat that underscores the robustness of the solution. Importantly, this was achieved without any flaring, exemplifying our commitment to environmental responsibility and operational excellence. The ability to manage stable flow rates across varying drawdowns facilitated the creation of an accurate inflow curve, deepening our understanding of the well's remaining deliverability potential. This breakthrough not only confirms the technology's practicality but also empowers operators to plan future field extensions more effectively and enhance oil recovery (EOR). The results solidify MPP boosting technology as a proven, reliable, and environmentally conscious solution that is paving the way for the future success of well testing and extended production, well boosting, and zero flaring, among other challenges that this technology can address.

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